

Zhibo He

(+44) 7544 594337 | zh825@ic.ac.uk | [LinkedIn](#) | [Google Scholar](#) | [ORCID](#) | [GitHub](#)

EDUCATION

• Imperial College London

Sep 2025 – Sep 2026 (expected)

MSc Security and Resilience: Science and Technology

London, UK

- Relevant courses: Software Reliability (COMP70024), Formal Methods for Safe AI (COMP70031)

• Xi'an Jiaotong-Liverpool University

Sep 2021 – Sep 2025

BSc Information Management and Information Systems

Suzhou, China

- Average: 77% (First Class Honours); Departmental Top 5%
- Academic Excellence Scholarship (2023–2024); Academic Achievement Scholarship (2022–2023)

RESEARCH INTERESTS

Program analysis, software testing, differential fuzzing, symbolic execution, software reliability, AI-assisted program reasoning, and LLMs for code and security.

SELECTED RESEARCH AND PROJECT EXPERIENCE

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MSc Thesis: Pre-discrepancy Frontiers in Differential Fuzzing

April 2026 – Present

Supervised by [Prof. Cristian Cadar](#)

- * Investigating whether parser-level diagnostic differences can guide differential fuzzing before observable output discrepancies appear.
- * Built a diagnostic-aware cJSON/parson differential fuzzing pipeline with AFL/AFL++, parser-native error positions, and normalized diagnostic classes.
- * Identified 74 stable diagnostic-frontier candidates from a 215-seed evaluation set and validated them across 10 reruns.
- * Designed lineage-based follow-up mutation experiments, scaling from repeated 100-mutation pilots to targeted 1000-mutation studies.
- * Analysed downstream discrepancies and found that stable candidates mainly correspond to invalid-JSON boundary cases, supporting a staged candidate-funnel view rather than a single universal precursor signal.

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Coursework Project: Greybox Fuzzer for SAT Solvers [\[GitHub\]](#)

Winter 2025

COMP70024 Software Reliability | Tools: C/C++, Python, gcov, ASan, UBSan

- * Built an AFL-like greybox mutation fuzzer for testing SAT solvers as part of the Software Reliability coursework.
- * Combined grammar-aware mutation with runtime feedback from gcov, AddressSanitizer, and UndefinedBehaviorSanitizer.
- * Designed mutation operators to improve coverage and expose solver robustness issues.
- * Implemented scripts for corpus mutation, execution monitoring, sanitizer-based testing, and coverage collection.

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BSc Thesis: Energy-Efficient Message Bundling for Wireless Sensor Networks

Spring 2025

Supervised by [Prof. Kyeong Soo Kim](#) | Tools: Python, PuLP, ILP, IEEE 802.15.4

- * Proposed REDS, a retransmission-aware, energy-efficient, and delay-constrained message bundling scheme for Wireless Sensor Networks.
- * Formulated an ILP-based optimization model incorporating end-to-end delay constraints, retransmission awareness, SNR, and Packet Error Rate.
- * Implemented numerical experiments in Python using PuLP to compute optimal bundling configurations under realistic IoT/WSN parameters.
- * Achieved 33.5% estimated network-wide energy saving, with upstream sensor nodes achieving up to 53.4% saving under the evaluated topology.

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Research Intern, Cybersecurity Laboratory, Geely Automotive Research Institute

Summer 2024

Topics: automotive cybersecurity, CAN bus robustness, OBD-II security evaluation

- * Investigated the practical effectiveness of academic CAN bus attack models on real-world vehicle systems.
- * Evaluated BUS-OFF attack feasibility in collaboration with researchers using CAN bus and OBD-II access scenarios.
- * Observed that gateway and intrusion-detection mechanisms mitigated OBD-II-based BUS-OFF attacks on modern connected vehicles.
- * Identified residual risk in legacy vehicles and direct-bus diagnostic settings, highlighting deployment-dependent security assumptions.

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Research Intern, Institute of Big Data, Fudan University

Summer 2023

Supervised by Prof. Ping Chen | Topics: LLM security, LLVM IR, code translation

- * Studied LLM security risks, including adversarial attacks, data leakage, and prompt-injection vulnerabilities.
- * Designed adversarial examples to evaluate the robustness of pretrained models under security-relevant perturbations.
- * Investigated LLM-assisted code translation using LLVM IR as an intermediate representation.
- * Compiled Java programs into LLVM IR and explored prompt-engineering strategies for translating LLVM IR into functionally equivalent Python code.

PUBLICATIONS

1. Zhai, S., **He, Z.**, Cong, X., Hou, J., Gui, J., You, J. W., Gong, X., Kwok, J. T.-Y., & Tang, Y. Y. (2025). **Unrevealed Threats: A Comprehensive Study of the Adversarial Robustness of Underwater Image Enhancement Models.** *IEEE Transactions on Multimedia*, 27, 9532–9541.

TECHNICAL SKILLS

- **Programming:** Python, C/C++, Java, Bash, SQL
- **Program Analysis & Testing:** AFL/AFL++, KLEE, fuzzing, differential testing, symbolic execution, parser instrumentation
- **Compilers & IR:** LLVM/Clang, LLVM IR, gcov
- **Systems & Security Tooling:** Linux, Git, Docker, AddressSanitizer, UndefinedBehaviorSanitizer
- **AI/LLM:** LLM security evaluation, prompt engineering, LLM-assisted code translation
- **IoT & Optimization:** Wireless Sensor Networks, IEEE 802.15.4, ILP modeling, PuLP
- **Databases:** Relational schema design, normalization, ER modeling
- **Languages:** English (IELTS 7.5)